

Notes: van Rooij, Lusardi, and Alessie (JFE, 2011)

Financial Literacy and Stock Market Participation

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1 Big picture

- **Question.** Does financial literacy affect whether households participate in the stock market?
- **Setting.** The paper uses two special modules added to the 2005–2006 De Nederlandsche Bank (DNB) Household Survey in the Netherlands to measure financial literacy and relate it to household stock ownership.
- **Core challenge.** Stock market participation and financial literacy are jointly endogenous. More financially sophisticated households may be more willing to hold stocks, but households may also learn by participating in the stock market. In addition, literacy measures are noisy because respondents sometimes guess or misunderstand the survey questions.
- **Main descriptive result.** Most respondents can answer simple numeracy and inflation questions, but advanced financial knowledge is much weaker. Many respondents do not know the difference between stocks and bonds, do not understand the bond price–interest rate relation, and are confused by diversification questions.
- **Main empirical result.** Households with greater advanced financial literacy are much more likely to own stocks or mutual funds. In the baseline OLS specification, a one-standard-deviation increase in advanced literacy is associated with roughly a 9 percentage point increase in stock market participation.
- **Main identification result.** Instrumenting advanced literacy with respondents’ exposure to siblings’ negative financial experiences and parents’ low financial understanding leaves the effect positive and significant, suggesting that financial literacy has an independent causal role.
- **Main interpretation.** Financial literacy matters because stocks are complex assets. Households with low literacy face higher information and participation costs, rely more on informal advice networks, and are less likely to enter the stock market.
- **Why this paper matters.** It is one of the classic papers linking household finance to directly measured financial knowledge rather than crude proxies such as education, income, or wealth.

2 Data and measurement (key objects)

2.1 Survey and stock market participation

- **Source.** The 2005–2006 De Nederlandsche Bank Household Survey (DHS), a representative household panel for the Netherlands.
- **Special modules.** The authors designed two literacy modules: one with standard literacy questions and one with additional questions on cognition, policy changes, and preferences.
- **Main literacy-module sample.** The literacy module contains about 1,508 respondents.
- **Unit of observation.** The household, typically linked to the respondent who answers the survey and reports household financial behavior.
- **Main financial outcome.** A dummy for stock market participation, defined as owning individual stocks and/or mutual funds.

- **Merged stock-ownership sample.** After merging the literacy module with finance and demographic information, the stock participation sample contains about 1,189 households; the main multivariate regressions use 1,115 observations.

2.2 Financial literacy questions and indices

- **Basic literacy questions.** Five simple questions on numeracy, interest compounding, inflation, time value of money, and money illusion.
- **Advanced literacy questions.** Eleven questions on the stock market, stocks, bonds, mutual funds, risk diversification, relative riskiness of assets, and the effect of interest rates on bond prices.
- **Randomized wording.** Three advanced questions are asked with alternative wording to test whether responses are robust to the phrasing of the question.
- **Basic literacy index.** Constructed by factor analysis on the five basic-question correctness indicators.
- **Advanced literacy index.** Constructed by factor analysis on dummies for correct and “do not know” answers to the advanced questions, with the factor analysis run separately across randomized wording groups.
- **Scoring method.** The paper uses the iterated principal factor method and then recovers factor scores using the Bartlett method.

2.3 Advice sources, demographics, and additional controls

- **Advice source.** Respondents report the most important source of advice for important household financial decisions: family/friends, newspapers, financial magazines/books, brochures, advertising, professional advisers, financial software, or the internet.
- **Demographics.** Education, age, gender, marital status, children, retirement status, self-employment, household income, and non-equity wealth.
- **Economics education.** A categorical variable describing how much of the respondent’s education was devoted to economics.
- **Use of economics in daily life.** A categorical variable capturing how much understanding of economics the respondent needs in work or other daily activities.
- **Self-assessed literacy.** Respondents rate their own understanding of economics on a 1–7 scale.
- **Alternative cognition proxies.** Difficulty with the euro cash changeover and difficulty understanding the Dutch health-insurance reform.
- **Risk aversion.** A set of hypothetical income-gamble questions similar to those in the HRS.
- **Peer variables.** Education and income of peers are used in robustness checks to see whether literacy is merely proxying for information transmitted through social networks.

2.4 Instruments

- **Sibling financial condition.** Whether the oldest sibling is in a better, the same, or a worse financial situation than the respondent.
- **Parents’ financial understanding.** Whether parents had high or low understanding of financial matters.
- **Economic idea.** Respondents may learn from the negative financial experiences of family members, so these variables shift financial literacy without directly being part of the stock participation decision.

2.5 Unobservables and timing

- **Main omitted-variable concern.** Financial literacy may proxy for cognitive ability, patience, risk tolerance, social networks, family background, or participation-driven learning.
- **Timing concern.** Literacy is observed at the same time as stock ownership, so reverse causality is a real issue.
- **Measurement concern.** Advanced literacy is noisy because many respondents answer “do not know” or appear to guess; the randomized wording experiment shows that some questions are sensitive to framing.

3 Models and empirical specifications (all equations + interpretation)

3.1 Constructing the literacy indices

The paper first summarizes the literacy questions into two latent indices. Conceptually, for the basic questions one can write

$$q_{ij}^B = \lambda_j^B F_i^B + u_{ij}, \quad j = 1, \dots, 5,$$

where q_{ij}^B is an indicator for whether respondent i answers basic question j correctly, F_i^B is latent basic literacy, and u_{ij} is an idiosyncratic component.

For the advanced questions, the paper uses both correct-answer and “do not know” indicators. Conceptually,

$$q_{ij}^{A,c} = \lambda_j^{A,c} F_i^A + u_{ij}^c, \quad q_{ij}^{A,d} = \lambda_j^{A,d} F_i^A + u_{ij}^d, \quad j = 1, \dots, 11,$$

where $q_{ij}^{A,c}$ is a correct-answer indicator, $q_{ij}^{A,d}$ is a “do not know” indicator, and F_i^A is latent advanced literacy.

- **Interpretation.** The factor-analysis step treats financial literacy as a latent object measured with error by a battery of survey questions.
- **Why the “do not know” answers matter.** Someone who says “do not know” is different from someone who confidently gives a wrong answer. The paper uses that extra information instead of collapsing all noncorrect answers together.

- **Why random wording matters.** Because some advanced questions are extremely sensitive to wording, the paper estimates the advanced factor separately across wording groups rather than pretending the items are perfectly comparable.

3.2 Baseline stock market participation regression

The main reduced-form specification is a linear probability model:

$$Stock_i = \alpha + \beta AdvLit_i + \gamma BasicLit_i + X_i'\delta + \varepsilon_i,$$

where $Stock_i$ is a dummy for stock market participation, $AdvLit_i$ is the advanced literacy index, $BasicLit_i$ is the basic literacy index, and X_i contains the large set of demographic and economic controls.

- **Interpretation of β .** It is the effect of advanced financial literacy on the probability of owning stocks or mutual funds, conditional on education, income, wealth, and many other observables.
- **Why linear probability?** The paper uses a simple and transparent specification and reports heteroskedasticity-robust standard errors.
- **Role of the basic literacy index.** It serves partly as a control for numeracy and cognitive ability, so the advanced literacy coefficient can be interpreted more narrowly as the effect of financial sophistication rather than raw arithmetic ability.

3.3 Why OLS may fail

The authors emphasize two distinct biases in OLS:

$$\hat{\beta}_{OLS} \neq \beta$$

if either of the following holds:

- **Measurement error.** The advanced literacy index is noisy, which tends to attenuate $\hat{\beta}_{OLS}$ downward.
- **Reverse causality / learning by doing.** Households that already participate in the stock market may accumulate literacy from experience, which biases $\hat{\beta}_{OLS}$ upward.
- **Interpretation.** The OLS result is informative but not enough to establish causality because the direction of bias is ambiguous.

3.4 First-stage equation for advanced literacy

To address endogeneity, the paper instruments advanced literacy with family-experience variables:

$$AdvLit_i = \pi_0 + Z_i'\pi_1 + X_i'\pi_2 + \nu_i,$$

where Z_i contains dummies for the oldest sibling's financial condition and parents' understanding of financial matters.

- **Interpretation.** Family background variables shift literacy through learning from the successes or failures of others.
- **Empirical sign pattern.** The first stage suggests that respondents become more financially literate when siblings are in worse financial condition or when parents had poorer financial understanding, which is more consistent with a learning channel than with a common family fixed effect.

3.5 Second-stage GMM equation

The structural second stage is

$$Stock_i = \alpha + \beta \widehat{AdvLit}_i + \gamma BasicLit_i + X_i' \delta + \varepsilon_i.$$

Because the outcome is binary and the error term is heteroskedastic, the paper implements the IV estimates with GMM rather than simple 2SLS.

- **Interpretation of β .** The coefficient now measures the effect of exogenous variation in advanced literacy induced by the family-based instruments.
- **Why GMM here?** In a linear probability model, heteroskedasticity is mechanical, so GMM is a natural implementation of IV with robust weighting.

3.6 Peer-effects extension

To test whether literacy merely proxies for social learning, the paper augments the baseline with peer education and income variables:

$$Stock_i = \alpha + \beta AdvLit_i + P_i' \eta + X_i' \delta + \varepsilon_i,$$

where P_i includes education or income of peers.

- **Interpretation.** If literacy only captures social exposure, then controlling for peer characteristics should sharply reduce β .
- **Result.** It does not.

3.7 Alternative cognition controls

The paper replaces the basic literacy index with direct proxies for ability to process complex economic changes:

$$Stock_i = \alpha + \beta AdvLit_i + C_i' \rho + X_i' \delta + \varepsilon_i,$$

where C_i are dummies for difficulty handling the euro introduction or the health-insurance reform.

- **Interpretation.** This checks whether advanced literacy is only standing in for general cognitive ability.
- **Result.** The literacy coefficient remains positive and significant.

3.8 Risk-aversion extension

Finally, the paper adds risk-aversion categories:

$$Stock_i = \alpha + \beta AdvLit_i + R_i'\theta + X_i'\delta + \varepsilon_i,$$

where R_i are indicators for degrees of risk tolerance.

- **Interpretation.** This checks whether advanced literacy is simply picking up risk preferences.
- **Result.** Again, the literacy effect remains strong.

3.9 Placebo and additional robustness

The authors also show that:

- literacy is *not* related to ownership of simple savings accounts,
- the results remain when excluding the most wording-sensitive advanced questions,
- the results remain when excluding self-employed households.
- **Interpretation.** The literacy effect is specific to complex financial participation rather than to all forms of saving.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- **Education is not enough.** Education is correlated with literacy, but it is an imperfect proxy. Even among respondents with a university degree, only 43.4% are in the top quartile of advanced literacy.
- **Cross-sectional correlation is not causation.** Literate households are richer, more educated, and more financially active, so raw differences in stock ownership could reflect many omitted variables.
- **Measurement error is serious.** The randomized wording experiments show that some “correct” advanced answers are unstable and may partly reflect guessing.

4.2 Why sibling financial condition helps identification

- The instrument is the relative financial situation of the oldest sibling.
- The key idea is that respondents may learn from family experiences that are not directly part of their own portfolio choice problem.
- The first-stage sign is informative: respondents with siblings in worse financial condition tend to have *higher* literacy, which supports the learning channel.
- If the instrument were mainly proxying for a favorable family background, one would have expected the opposite sign.

4.3 Why parents' financial understanding helps identification

- Parents' understanding of financial matters also shifts literacy through family transmission and negative learning.
- Respondents with parents who had poor financial understanding are more likely to score higher on advanced literacy, again pointing to corrective learning rather than inherited advantage.
- Taken together, the sibling and parent variables provide overidentification, which allows the paper to check the plausibility of the exclusion restriction.

4.4 Instrument strength and specification tests

- **First-stage strength.** The first-stage F-statistics are comfortably above conventional weak-instrument concerns in the main IV specifications (11.52 with sibling instruments only; 8.82 with siblings plus parents).
- **Hansen J-test.** The overidentifying restrictions are not rejected in the main two-instrument-plus-parent specification.
- **Exogeneity test.** The paper does not reject exogeneity of the advanced literacy variable, but the IV estimates are nevertheless useful because they address the two countervailing biases that plague OLS.

4.5 What the IV design does and does not do

- **What it does.** It isolates variation in literacy coming from family experience rather than from direct stock-market participation.
- **What it does not do.** It does not fully solve every possible family-background concern. The authors are transparent that these instruments could still be debated, which is why they complement the IV with many robustness checks.

4.6 Relation to the literature

- The paper speaks to the literature on household portfolio choice and the stock market participation puzzle.
- It also connects to the literature on social capital, peer effects, and informal financial networks.
- Relative to earlier work that used education, income, or wealth as proxies for sophistication, this paper measures financial literacy directly and then studies whether it has predictive power above those traditional variables.

5 Tables and figures

5.1 Table 2: Advanced financial literacy

Read:

- Advanced literacy is low and uneven. The mean respondent answers only 5.93 of the 11 advanced questions correctly, with 2.33 incorrect and 2.65 “do not know” answers on average.
- Knowledge is especially weak on sophisticated concepts. Only 24.6% correctly answer the bond-price question, and only 30.0% correctly reject the statement that a 10-year bond cannot be sold after 5 years without a major penalty.
- Even in a finance-rich country sample, only 5.0% answer all 11 advanced questions correctly.
- The main message is that basic numeracy is common, but true market knowledge is scarce.

Advanced financial literacy

Panel A reports the proportion of households providing correct, incorrect, and “do not know” answers to each of the 11 advanced literacy questions. Panel B reports the distribution of the number of correct, incorrect, and “do not know” answers to the 11 advanced literacy questions. The data are from the 2005–2006 2006 Household Survey.

Panel A: Weighted percentages of total number of respondents (N=1,508)

	Correct	Incorrect	Do not know
Which statement describes the main function of the stock market? ¹⁾	67.0	12.9	19.7
What happens if somebody buys the stock of firm B in the stock market? ¹⁾	62.2	20.6	17.0
Which statement about mutual funds is correct? ¹⁾	66.7	11.2	21.7
What happens if somebody buys a bond of firm A? ¹⁾	51.5	17.7	28.4
Considering a long time period (for example 10 or 20 years), which asset normally gives the highest return: savings accounts, bonds, or stocks? ¹⁾	47.2	30.1	22.3
Normally, which asset displays the highest fluctuations over time: savings accounts, bonds, or stocks? ¹⁾	68.5	12.7	18.4
When an investor spends his money among different assets, does the risk of losing money increase, decrease, or stay the same? ¹⁾	63.3	17.4	19.0
If you have a 10-year bond, it means you cannot sell it after 5 years without incurring a major penalty. True or false? ²⁾	30.0	38.1	31.9
Stocks are normally riskier than bonds. True or false? ²⁾	60.2	15.1	24.3
Buying a company fund usually provides a safer return than a stock mutual fund. True or false? ²⁾	48.2	34.8	28.5
If the interest rate falls, what should happen to bond prices: rise/fall/stay the same/none of the above? ²⁾	24.6	37.1	37.5

¹⁾ See exact wording in Section 3 of the test.
²⁾ This question has been phrased in two different ways. See also Table 3.
Note: Correct, incorrect, and do not know responses do not sum up to 100% because of refusals.

Panel B: Summary of responses

Weighted percentages of total number of respondents (N=1,508)

	Number of correct, incorrect, and do not know answers (out of eleven questions)											All	Mean
	None	1	2	3	4	5	6	7	8	9	10		
Correct	7.6	5.1	5.2	6.4	7.3	10.0	11.1	11.3	10.8	9.8	5.0	5.93	
Incorrect	18.7	20.2	19.8	18.8	16.4	7.1	4.7	1.6	0.6	0.1	0.0	2.33	
Do not know	48.2	11.4	8.0	6.1	5.1	3.7	4.1	4.2	2.8	3.2	9.5	2.65	

Note: Categories do not sum up to 100% and means do not sum up to 11 due to rounding and refusals.

Table 3
Advanced literacy: Responses to questions with inverted wording.
This table reports the proportion of households providing correct, incorrect, and “do not know” answers to the three advanced literacy questions that were asked in two versions with a slightly different wording. For each of these three questions, households were assigned randomly to one of the two versions. The table shows weighted percentages and the Pearson chi-square statistic to test the null hypothesis that the wording of the question is irrelevant for the distribution of correct and incorrect answers (p-values reported in parentheses). The data are from the 2005–2006 Household Survey

	Correct	Incorrect	Do not know
Stocks are normally riskier than bonds. True or false? (N=751)	60.8	17.1	21.7
Bonds are normally riskier than stocks. True or false? (N=757)	59.7	13.1	26.9
Pearson chi2(2)=5.25 (p=0.072)			
Buying a company stock usually provides a safer return than a stock mutual fund. True or false? (N=763)	63.4	12.1	24.1
Buying a stock mutual fund usually provides a safer return than a company stock. True or false? (N=745)	32.3	38.1	29.2
Pearson chi2(2)=184.59 (p=0.000)			
If the interest rate falls, what should happen to bond prices: rise/fall/stay the same/none of the above? (N=755)	30.5	33.8	34.8
If the interest rate rises, what should happen to bond prices: rise/fall/stay the same/none of the above? (N=753)	18.9	40.3	40.3
Pearson chi2(2)=23.15 (p=0.000)			

Note: Correct, incorrect, and do not know responses do not sum up to 100% because of refusals. In performing the test, we group together “do not know” and “refusals.”

5.2 Table 3: Responses to questions with inverted wording

Read:

- Question wording matters a lot for advanced literacy.
- For the mutual-fund-versus-company-stock question, the share of correct answers changes from 63.4% under one wording to 32.3% under the inverted wording, with a highly significant chi-square test.
- For the bond-price question, the correct share changes from 30.5% to 18.9% depending on wording.
- This table is crucial because it shows that advanced literacy measures are noisy. The paper therefore treats literacy as a latent variable and uses factor analysis rather than reading each response literally.

5.3 Table 4: Literacy across demographics

Read:

- Financial literacy rises strongly with education, but education is not the same thing as literacy.
- Only 43.4% of respondents with a university degree are in the top quartile of advanced literacy, versus 70.9% for the top quartile of *basic* literacy.
- Gender gaps are large: 34.5% of women are in the bottom quartile of advanced literacy, compared with 15.9% of men; only 12.1% of women are in the top quartile, versus 37.2% of men.
- These patterns motivate the paper’s insistence that education alone is too crude to stand in for financial sophistication.

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Table 5
Most important source of advice for different levels of literacy.
This table shows, for different levels of basic and advanced financial literacy, the sources of advice used in making important financial decisions. We group the basic and advanced literacy measures in four quartiles and report weighted percentages of households within each literacy quartile using a specific source of information (N=1,135). The data are from the 2005–2006 DNB Household Survey.

What is your most important source of advice when you have to make important financial decisions for the household?	Basic literacy quartiles			
	1 (low)	2	3	4 (high)
• Parents, friends, or acquaintances	40.2	34.4	28.8	20.8
• Information from the newspapers	3.6	7.8	8.9	9.5
• Financial magazines, guides, books	3.9	7.5	9.3	12.4
• Brochures from my bank or mortgage adviser	10.6	6.8	6.0	8.1
• Advertisements on TV, in papers, or other media	3.7	3.2	2.8	3.9
• Professional financial advisers	21.8	21.3	24.2	25.5
• Financial computer programs	0.0	0.3	0.9	0.7
• Financial information on the Internet	4.0	7.5	8.1	10.5
• Other	12.3	11.4	11.0	8.6

What is your most important source of advice when you have to make important financial decisions for the household?	Advanced literacy quartiles			
	1 (low)	2	3	4 (high)
• Parents, friends, or acquaintances	40.7	37.4	19.9	17.9
• Information from the newspapers	1.1	6.0	10.6	13.7
• Financial magazines, guides, books	2.1	7.6	9.7	17.0
• Brochures from my bank or mortgage adviser	6.6	6.7	11.3	6.2
• Advertisements on TV, in papers, or other media	4.0	3.6	5.0	1.4
• Professional financial advisers	19.4	23.6	27.5	24.1
• Financial computer programs	0.2	0.3	1.1	0.5
• Financial information on the Internet	6.3	6.6	7.6	12.4
• Other	19.7	8.2	7.3	6.9

Note: Percentages may not sum up to 100 due to rounding.

5.4 Table 5: Most important source of advice

Read:

- Low-literacy households rely much more on informal sources of advice.
- In the bottom quartile of advanced literacy, 40.7% cite parents, friends, or acquaintances as the main advice source; in the top quartile, that falls to 17.9%.
- High-literacy households are more likely to rely on newspapers, financial magazines, the internet, and professional advisers.
- The paper uses this table to suggest a mechanism: literacy changes how households gather and process financial information.

5.5 Table 6: Stock market participation across subgroups

Read:

- Overall stock market participation is only 23.8%.
- Participation rises sharply with advanced literacy: 7.5% in the bottom quartile, 15.0% in the second quartile, 26.5% in the third quartile, and 44.4% in the top quartile.
- Participation also rises with education, income, wealth, age, and being male.
- This table gives the central descriptive fact of the paper before moving to multivariate regressions.

5.6 Table 7: Multivariate OLS results

Read:

Table 6
Stock market participation across subgroups

This table shows stock ownership across different levels of education and different age groups, across gender and marital status, and across different levels of net household income, wealth (excluding equity wealth), and basic as well as advanced literacy. Income, wealth, and literacy are grouped in four quartiles. The table reports weighted percentages and stock ownership is defined as owning individual stocks and/or mutual funds ($N=1,189$). The data are from the 2005–2006 DNB Household Survey.

Education		Age	
Primary	11.3	21–30 years	14.4
Preparatory intermediate voc.	16.0	31–40 years	19.4
Intermediate vocational	19.1	41–50 years	27.1
Secondary pre-university	22.5	51–60 years	26.8
Higher vocational	33.7	61–70 years	24.3
University	38.8	71 years and older	30.1
Gender		Marital status	
Female	16.7	Not married	19.8
Male	30.3	Married	26.8
Net household income quartiles		Non-equity net wealth quartiles	
1 (low)	13.4	1 (low)	7.1
2	17.5	2	20.3
3	29.1	3	29.7
4 (high)	35.9	4 (high)	37.9
Basic literacy quartiles		Advanced literacy quartiles	
1 (low)	7.7	1 (low)	7.5
2	21.2	2	15.0
3	22.0	3	26.5
4 (high)	32.8	4 (high)	44.4

Table 7
Multivariate analysis of stock market participation: OLS results.

This table reports OLS estimates of the effect of financial literacy and several control variables on stock market participation. The data are from the 2005–2006 DNB Household Survey.

	OLS (1)		OLS (2)		OLS (3)	
Advanced literacy index			0.0892***	(0.0118)	0.0756***	(0.0125)
Age dummies (Base group: age ≤ 30)						
30 < age ≤ 40	–0.0250	(0.0456)	–0.00850	(0.0452)	–0.0119	(0.0451)
40 < age ≤ 50	0.0261	(0.0469)	0.0353	(0.0467)	0.0335	(0.0464)
50 < age ≤ 60	0.0133	(0.0489)	0.0165	(0.0474)	0.0202	(0.0474)
Age > 60	0.0504	(0.0617)	0.0734	(0.0595)	0.0888	(0.0597)
Education dummies (Base group: < int. vocational)						
Intermediate vocational	0.0352	(0.0372)	0.0247	(0.0363)	0.0279	(0.0365)
Secondary pre-university	0.0700*	(0.0411)	0.0298	(0.0414)	0.0242	(0.0427)
Higher vocational	0.110***	(0.0369)	0.0717*	(0.0366)	0.0651*	(0.0372)
University	0.153***	(0.0467)	0.102**	(0.0472)	0.0969**	(0.0471)
Male	0.106***	(0.0277)	0.0715***	(0.0275)	0.0591**	(0.0278)
Married	–0.0367	(0.0315)	–0.0267	(0.0312)	–0.0302	(0.0311)
Number of children	–0.00159	(0.0154)	0.00290	(0.0151)	0.00247	(0.0150)
Retired	–0.0252	(0.0550)	–0.0111	(0.0530)	–0.0305	(0.0531)
Self-employed	0.0458	(0.0557)	0.0319	(0.0574)	0.0206	(0.0579)
Ln(household income)	0.0916***	(0.0270)	0.0648***	(0.0263)	0.0813***	(0.0260)
Second wealth quartile ($\leq 2,300 < \text{wealth} \leq \leq 45,500$)	0.100***	(0.0353)	0.0749**	(0.0346)	0.0834***	(0.0345)
Third wealth quartile ($\leq 45,500 < \text{wealth} \leq \leq 197,300$)	0.155***	(0.0368)	0.117***	(0.0366)	0.124***	(0.0364)
Fourth wealth quartile ($\text{wealth} > \leq 197,300$)	0.212***	(0.0419)	0.160***	(0.0425)	0.162***	(0.0422)
Basic literacy index					0.0147	(0.0104)
Economics education dummies (Base group: A lot)						
Some					–0.0200	(0.0418)
Hardly at all					–0.0117	(0.0456)
Daily use of economics dummies (Base group: A lot)						
Some					–0.103***	(0.0442)
Little					–0.142***	(0.0480)
Hardly at all					–0.0536	(0.0628)
Constant	–0.886***	(0.258)	–0.760***	(0.251)	–0.668**	(0.255)
Observations		1115		1115		1115
R-squared		0.103		0.135		0.146
p-Value test age coefficients=0		0.613		0.598		0.448
p-Value test education coefficients=0		0.0343		0.163		0.240
p-Value test economics education coefficients=0						0.931
p-Value test daily use of economics coefficients=0						0.0183

Notes: Robust standard errors are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

- Adding advanced literacy to the baseline linear probability model yields a coefficient of 0.0892 (standard error 0.0118), highly significant.
- After adding the basic literacy index, economics education, and daily use of economics, the advanced literacy coefficient remains large at 0.0756 (standard error 0.0125).
- The basic literacy coefficient is small and not statistically significant once advanced literacy is included.
- The core message is that advanced financial sophistication predicts stock market participation above and beyond education, income, wealth, and many standard controls.

5.7 Table 8A and Table 8B: First stage and GMM estimates

Table 8A
First-stage regressions.

This table shows the estimates of the first-stage regressions of advanced financial knowledge on the set of controls and dummy variables indicating the financial situation of the oldest sibling in comparison to the financial situation of the respondent. The reference group consists of those respondents whose oldest sibling is in a better financial situation. In addition to these variables, in the second column, two dummies for the parents' understanding of financial matters are added. The reference group consists of those respondents whose parents have the lowest understanding of financial matters. The data are from the 2005–2006 DNB Household Survey.

	OLS (1)		OLS (2)	
Age dummies (Base group: age ≤ 30)				
30 < age ≤ 40	–0.152	(0.111)	–0.155	(0.112)
40 < age ≤ 50	–0.0987	(0.106)	–0.101	(0.107)
50 < age ≤ 60	0.0304	(0.110)	0.0167	(0.110)
Age > 60	0.000778	(0.140)	0.00373	(0.140)
Education dummies (Base group < intermed. vocational)				
Intermediate vocational	0.106	(0.0893)	0.0891	(0.0885)
Secondary pre-university	0.287***	(0.0871)	0.270***	(0.0873)
Higher vocational	0.212***	(0.0741)	0.197***	(0.0734)
University	0.341***	(0.0836)	0.332***	(0.0833)
Male	0.281***	(0.0607)	0.292***	(0.0607)
Married	–0.129*	(0.0678)	–0.119*	(0.0677)
Number of children	–0.0278	(0.0297)	–0.0297	(0.0298)
Retired	0.0649	(0.108)	0.0490	(0.108)
Self-employed	0.0732	(0.0947)	0.0706	(0.0922)
Ln(household income)	0.0614	(0.0564)	0.0436	(0.0560)
Second wealth quartile ($\leq 2,300 < \text{wealth} \leq \leq 45,500$)	0.260***	(0.0982)	0.256***	(0.0976)
Third wealth quartile ($\leq 45,500 < \text{wealth} \leq \leq 197,300$)	0.377***	(0.0945)	0.364***	(0.0938)
Fourth wealth quartile ($\text{wealth} > \leq 197,300$)	0.447***	(0.0996)	0.438***	(0.0999)
Basic literacy index	0.258***	(0.0298)	0.255***	(0.0295)
Economics education dummies (Base group: A lot)				
Some	–0.152**	(0.0634)	–0.163**	(0.0635)
Little	–0.214***	(0.0739)	–0.227***	(0.0739)
Hardly at all	–0.452***	(0.0876)	–0.468***	(0.0882)
Daily use of economics dummies (Base group: A lot)				
Some	–0.149**	(0.0754)	–0.146*	(0.0747)
Little	–0.299***	(0.0845)	–0.294***	(0.0842)
Hardly at all	–0.273*	(0.143)	–0.272*	(0.143)
Financial situation oldest sibling (Base group: no sibling)				
Worse	0.385***	(0.0889)	0.308***	(0.0877)
The same or better	0.184**	(0.0838)	0.113	(0.0827)
Parents' understanding of financial matters (Base: low)				
Intermediate or high				
Don't know				
Constant	–0.881	(0.557)	–0.381	(0.562)
Observations		1035		1035
R-squared		0.332		0.342
F-statistic first-stage regression		11.52		8.818
p-Value test instruments=0		1.13e–05		5.35e–07
p-Value test economics education coefficients=0		0.272		0.314
p-Value test age coefficients=0		0.000456		0.000091
p-Value test education coefficients=0		5.19e–06		2.40e–06
p-Value test economics education coefficients=0		0.00408		0.00459
p-Value test daily use of economics coefficients=0		8.02e–05		0.000138

Notes: Robust standard errors are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8B
Multivariate analysis of stock market participation: GMM results.

This table reports GMM estimates of the effect of financial literacy and several control variables on stock market participation. The advanced literacy index has been instrumented using two dummy variables indicating the financial situation of the oldest sibling (GMM (1)). The reference group consists of those respondents whose oldest sibling is in a better financial situation. Additionally, two dummies measuring parents' understanding of financial matters are used as instruments; the reference group is the respondents whose parents have the lowest understanding of financial matters (GMM (2)). The data are from the 2005–2006 Household Survey.

	GMM (1)		GMM (2)	
Advanced literacy index	0.216*	(0.124)	0.177**	(0.0861)
Age dummies (Base group: age ≤ 30)				
30 < age ≤ 40	0.0171	(0.0541)	0.0106	(0.0507)
40 < age ≤ 50	0.0724	(0.0530)	0.0686	(0.0508)
50 < age ≤ 60	0.0369	(0.0520)	0.0380	(0.0508)
Age > 60	0.0971	(0.0619)	0.0965	(0.0609)
Education dummies (Base group < intermed. vocational)				
Intermediate vocational	0.0109	(0.0410)	0.0150	(0.0392)
Secondary pre-university	–0.00391	(0.0579)	0.00663	(0.0517)
Higher vocational	0.0471	(0.0475)	0.0551	(0.0411)
University	0.0413	(0.0666)	0.0557	(0.0576)
Male	0.0190	(0.0435)	0.0303	(0.0367)
Married	–0.0316	(0.0372)	–0.0371	(0.0347)
Number of children	0.0133	(0.0163)	0.0125	(0.0159)
Retired	–0.00871	(0.0544)	–0.00625	(0.0538)
Self-employed	0.0117	(0.0640)	0.0136	(0.0631)
Ln(household income)	0.0737**	(0.0292)	0.0764*	(0.0279)
Second wealth quartile ($\leq 2,300 < \text{wealth} \leq \leq 45,500$)	0.0398	(0.0508)	0.0495	(0.0449)
Third wealth quartile ($\leq 45,500 < \text{wealth} \leq \leq 197,300$)	0.0675	(0.0529)	0.0820	(0.0523)
Fourth wealth quartile ($\text{wealth} > \leq 197,300$)	0.0924	(0.0755)	0.111*	(0.0621)
Basic literacy index	–0.0210	(0.0536)	–0.0105	(0.0265)
Economics education dummies (Base group: A lot)				
Some	0.0224	(0.0467)	0.0170	(0.0448)
Little	0.0324	(0.0540)	0.0238	(0.0497)
Hardly at all	0.0428	(0.0759)	0.0240	(0.0628)
Daily use of economics dummies (Base group: A lot)				
Some	–0.0742	(0.0489)	–0.0803*	(0.0469)
Little	–0.0898	(0.0612)	–0.102*	(0.0552)
Hardly at all	–0.0258	(0.0716)	–0.0349	(0.0678)
Constant	–0.558*	(0.296)	–0.587*	(0.275)
Observations		1035		1035
R-squared		0.090		0.120
p-Value test age coefficients=0		0.387		0.367
p-Value test education coefficients=0		0.780		0.678
p-Value test economics education coefficients=0		0.943		0.972
p-Value test daily use of economics coefficients=0		0.324		0.207
Hansen J test p-value		0.0663		0.295
F-statistic first-stage regression		11.52		8.818
p-Value exogeneity test		0.299		0.255

Notes: Robust standard errors are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Read:

- The family-based instruments are relevant in the first stage, with strong enough F-statistics to support IV estimation.
- In the second stage, the GMM coefficient on advanced literacy is 0.216 (standard error 0.124) when using sibling instruments only and 0.177 (standard error 0.0861) when adding the parent instruments.
- These coefficients are larger than the OLS estimates, which is consistent with attenuation from measurement error dominating any upward bias from reverse causality.
- The Hansen J-test and exogeneity tests do not undermine the specification, so the IV evidence strengthens the interpretation that literacy matters causally.

5.8 Table 9: Peer effects

Read:

- The paper asks whether literacy is just proxying for social interactions.
- Controlling for the education of peers leaves the advanced literacy coefficient essentially intact: OLS is 0.0828 and the GMM estimates remain around 0.19–0.20.
- Controlling for peer income also leaves the result strong, with OLS at 0.0850 and GMM estimates around 0.23–0.28.
- Hence literacy matters above and beyond peer composition.

Table 9
Stock market participation and the importance of peer effects.

This table reports OLS and GMM estimates of the effect of literacy on stock market participation, controlling for the education of peers. The advanced literacy index has been instrumented with two dummy variables indicating the financial situation of the oldest sibling (GMM (1)). The reference group consists of those respondents whose oldest sibling is in better financial condition. Additionally, two dummies measuring parents' understanding of financial matters are used as instruments; the reference group is the respondents whose parents have the lowest understanding of financial matters (GMM (2)). The data are from the 2005–2006 DNB Household Survey.

Panel A: Education of peers	OLS	GMM (1)	GMM (2)
Advanced literacy index	0.0828*** (0.0127)	0.202 (0.125)	0.187** (0.0828)
Education of peers (Base group: primary)			
(Pre-) intermediate vocational and sec. pre-university	0.0587 (0.0569)	0.0194 (0.103)	0.0221 (0.0679)
Higher vocational and university	0.135*** (0.0569)	0.103 (0.0679)	0.106* (0.0626)
Demographics and other control variables (see Table 7)	Yes	Yes	Yes
Observations	1044	979	979
R-squared	0.165	0.132	0.143
p-Value test education coefficients=0	0.900	0.879	0.877
p-Value test education peers coefficients=0	0.0351	0.0728	0.0608
Hansen J-test p-value		0.0796	0.358
F-statistic first-stage regression		10.10	8.415
p-Value exogeneity test		0.383	0.212
Panel B: Income of peers	OLS	GMM (1)	GMM (2)
Advanced literacy index	0.0850*** (0.0130)	0.280* (0.146)	0.229** (0.0959)
Income of peers (Base group: income < €20,000)			
€20,000 ≤ income < €28,000	−0.0242 (0.0450)	−0.0769 (0.0584)	−0.0643 (0.0505)
€28,000 ≤ income < €38,000	0.0435 (0.0481)	0.0298 (0.0525)	0.0337 (0.0504)
Income > = €38,000	0.0565 (0.0506)	0.0522 (0.0554)	0.0574 (0.0530)
Do not know	0.0150 (0.0382)	0.0769 (0.0556)	0.0653 (0.0471)
Demographics and other control variables (see Table 7)	Yes	Yes	Yes
Observations	1044	979	979
R-squared	0.163	0.059	0.113
p-Value test income peers coefficients=0	0.488	0.165	0.124
Hansen J-test p-value		0.132	0.457
F-statistic first-stage regression		8.802	6.786
p-Value exogeneity test		0.214	0.140

Note: Robust standard errors in parentheses; ***p < 0.01, **p < 0.05 *p < 0.1.

Table 10
Stock market participation and alternative measures of basic literacy.

This table reports OLS and GMM estimates of the effect of financial literacy on stock market participation, using two different controls for ability: the difficulty faced with the currency transition from the Dutch guilder to the euro and with the introduction of a new health care system, respectively. In Panel A, the reference group consists of those respondents who found dealing with the euro transition "very difficult" or who answered the question with "do not know (DK)". In Panel B, the reference group consists of those respondents who report no difficulty understanding the health care system change (see question H1 in Appendix B). The three dummy variables are based on question H2 in Appendix B. The advanced literacy index has been instrumented using two dummy variables indicating the financial situation of the oldest sibling (GMM (1)). The reference group consists of those respondents whose oldest sibling is in better financial condition. Additionally, two dummies measuring parents' understanding of financial matters are used as instruments; the reference group is respondents whose parents have the lowest understanding of financial matters (GMM (2)). The data are from the 2005–2006 DNB Household Survey.

Panel A: Euro introduction	OLS	GMM (1)	GMM (2)
Advanced literacy index	0.0801*** (0.0123)	0.166* (0.0893)	0.157** (0.0664)
Dealing with euro (Base group: very difficult or DK)			
Somewhat difficult	−0.0466 (0.0452)	−0.0377 (0.0472)	−0.0370 (0.0468)
Not very difficult	−0.0108 (0.0417)	−0.0129 (0.0447)	−0.0118 (0.0438)
Not difficult at all	0.0430 (0.0487)	0.0399 (0.0523)	0.0404 (0.0509)
Demographics and other control variables (see Table 7)	Yes	Yes	Yes
Observations	1053	1035	1035
R-squared	0.148	0.130	0.135
p-Value test euro coefficients=0	0.190	0.335	0.317
Hansen J-test p-value		0.0685	0.334
F-statistic first-stage regression		17.51	11.56
p-Value exogeneity test		0.370	0.259
Panel B: Change health insurance system	OLS	GMM (1)	GMM (2)
Advanced literacy index	0.0829*** (0.0126)	0.179** (0.0892)	0.167** (0.0677)
Difficulty health care system (Base group: not difficult)			
Making comparisons and collecting information	−0.00787 (0.0305)	−0.00291 (0.0312)	−0.00389 (0.0309)
Figuring out what the best is for me to do	−0.0230 (0.0365)	−0.00777 (0.0410)	−0.00984 (0.0398)
I don't know how to make these decisions & DK	0.0831 (0.0738)	0.0734 (0.0805)	0.0685 (0.0778)
Demographics and other control variables (see Table 7)	Yes	Yes	Yes
Observations	1053	1035	1035
R-squared	0.145	0.119	0.127
p-Value test health insurance coefficients=0	0.562	0.768	0.779
Hansen J-test p-value		0.0622	0.306
F-statistic first-stage regression		18.18	11.85
p-Value exogeneity test		0.304	0.221

Note: Robust standard errors in parentheses; ***p < 0.01, **p < 0.05 *p < 0.1.

5.9 Table 10: Alternative ability measures

Read:

- Replacing the basic literacy index with direct proxies for cognitive difficulty does not eliminate the literacy effect.
- With the euro-transition controls, the OLS coefficient on advanced literacy is 0.0801 and the GMM coefficients are 0.166 and 0.157.

Table 11

Stock market participation, literacy, and risk aversion.

This table reports OLS and GMM estimates of the effect of literacy on stock market participation, controlling for risk aversion. In this regression, the reference group consists of those respondents who exhibit the highest degree of risk aversion according to the questions reported in Appendix 8. The advanced literacy index has been instrumented using two dummy variables indicating the financial situation of the oldest sibling (GMM (1)). The reference group consists of those respondents whose oldest sibling is in better financial condition. Additionally, two dummies measuring parents' understanding of financial matters are used as instruments with the respondents whose parents have the lowest understanding of financial matters (GMM (2)) as a reference group. The data are from the 2005–2006 DNB Household Survey.

	OLS		GMM (1)		GMM (2)	
Advanced literacy index	0.0915***	(0.0145)	0.233*	(0.130)	0.188**	(0.0897)
Risk aversion (Base group: highest degree of aversion)						
Low	0.0555	(0.0458)	0.0306	(0.0534)	0.0398	(0.0489)
Medium	0.000791	(0.0547)	−0.0437	(0.0713)	−0.0292	(0.0631)
High	−0.0504	(0.0834)	−0.0768	(0.100)	−0.0618	(0.0939)
Don't know	0.00977	(0.0637)	0.0388	(0.0737)	0.0309	(0.0687)
Demographics and other control variables (see Table 7)	Yes	Yes	Yes	Yes	Yes	Yes
Observations		888		837		837
R-squared		0.138		0.090		0.120
p-Value test risk aversion coefficients=0		0.736		0.741		0.762
Hansen J-test p-value				0.0204		0.0879
F-statistic first-stage regression				12.72		9.114
p-Value exogeneity test				0.324		0.289

Note: Robust standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

- With the health-insurance-reform controls, the OLS coefficient is 0.0829 and the GMM coefficients are 0.179 and 0.167.
- So the paper's literacy measure is not merely standing in for a general inability to process policy changes or perform practical calculations.

5.10 Table 11: Risk aversion

Read:

- Adding risk-aversion controls also leaves the literacy coefficient large: OLS is 0.0915, while the GMM estimates are 0.233 and 0.188.
- The risk-aversion dummies themselves do not absorb the main literacy effect.
- This matters because a standard alternative explanation for low stock participation is simply high risk aversion.
- The table implies that the paper is capturing something distinct: knowledge, sophistication, and lower participation costs.

6 Short summary

- **Question.** Why do so few households participate in the stock market, and does financial literacy help explain this fact?
- **Answer.** Yes. Financial literacy, especially advanced literacy about financial markets and products, is strongly associated with stock market participation and remains important after controlling for education, income, wealth, cognition, peer effects, and risk aversion.
- **Identification.** The paper combines direct measurement of literacy, randomized wording checks, and IV estimation using family-based instruments.
- **Economic lesson.** Stocks are complicated assets, and households with low financial sophistication face real information frictions that keep them out of the market.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that financial literacy is not just a descriptive correlate of stock ownership; it is an economically important determinant of whether households enter the stock market.
- It also establishes that education is too blunt a measure of sophistication. Even highly educated households can have low advanced financial literacy.
- Finally, it shows that the content and presentation of financial questions matter for measurement, so empirical work should treat literacy as a noisy latent construct rather than as a set of perfectly measured test scores.

7.2 Economic intuition

- Basic saving products, such as savings accounts, require little specialized knowledge. Stocks and mutual funds do not. They require households to understand risk, diversification, and the functioning of financial markets.
- When households lack that knowledge, the fixed cost of participation is higher. They rely more on informal advice, are less able to evaluate opportunities, and are more likely to stay out of the market.
- Advanced financial literacy therefore works like a reduction in information and decision costs.

7.3 Takeaway

This paper is a benchmark in household finance because it shows, with direct survey measurement and a careful identification strategy, that financial sophistication matters for real financial behavior. The broad lesson is simple but powerful: participation in financial markets depends not only on income, wealth, and preferences, but also on whether households understand the products and institutions they face. Any theory of household portfolio choice that ignores financial literacy is likely to miss a key part of the stock market participation puzzle.